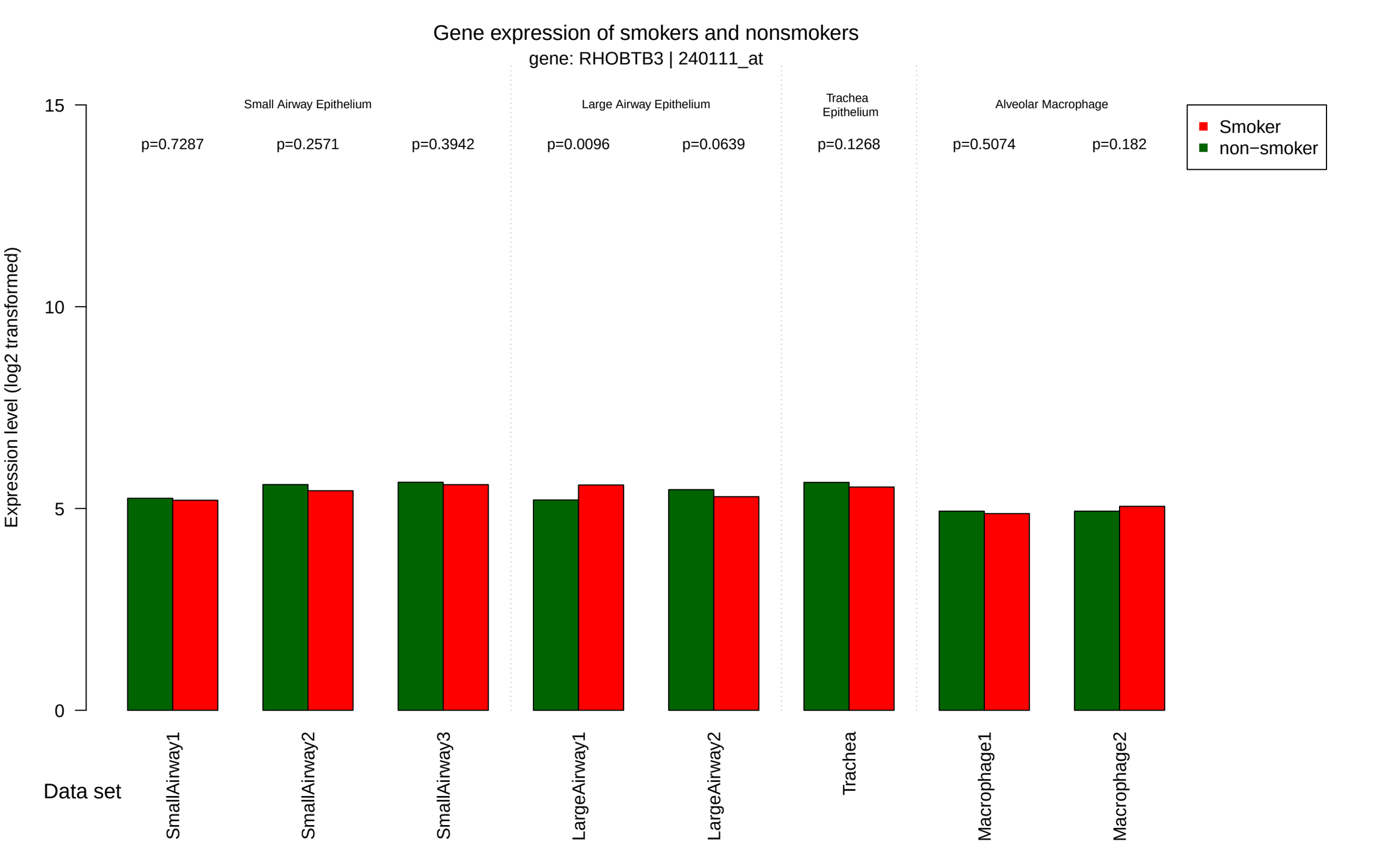
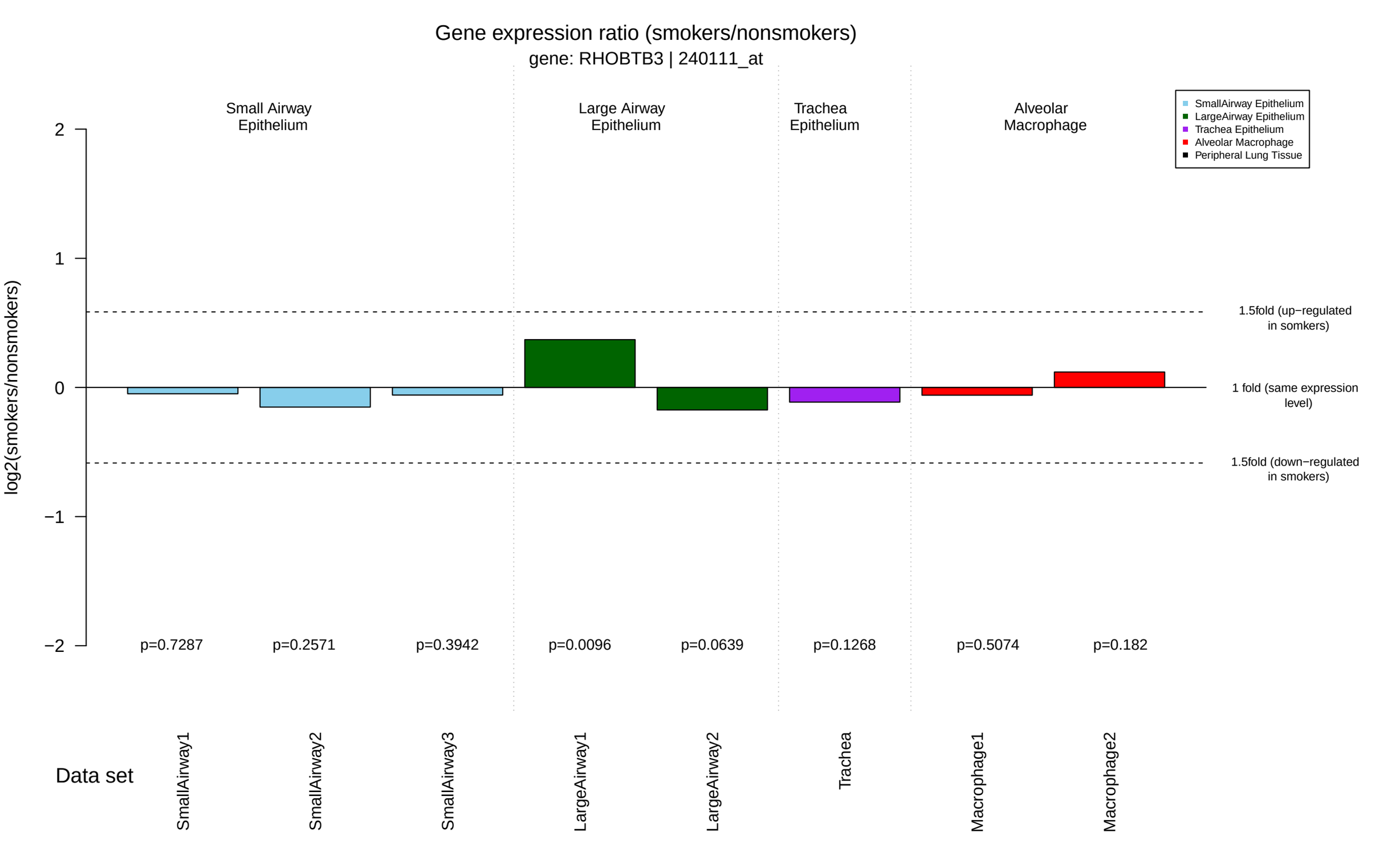
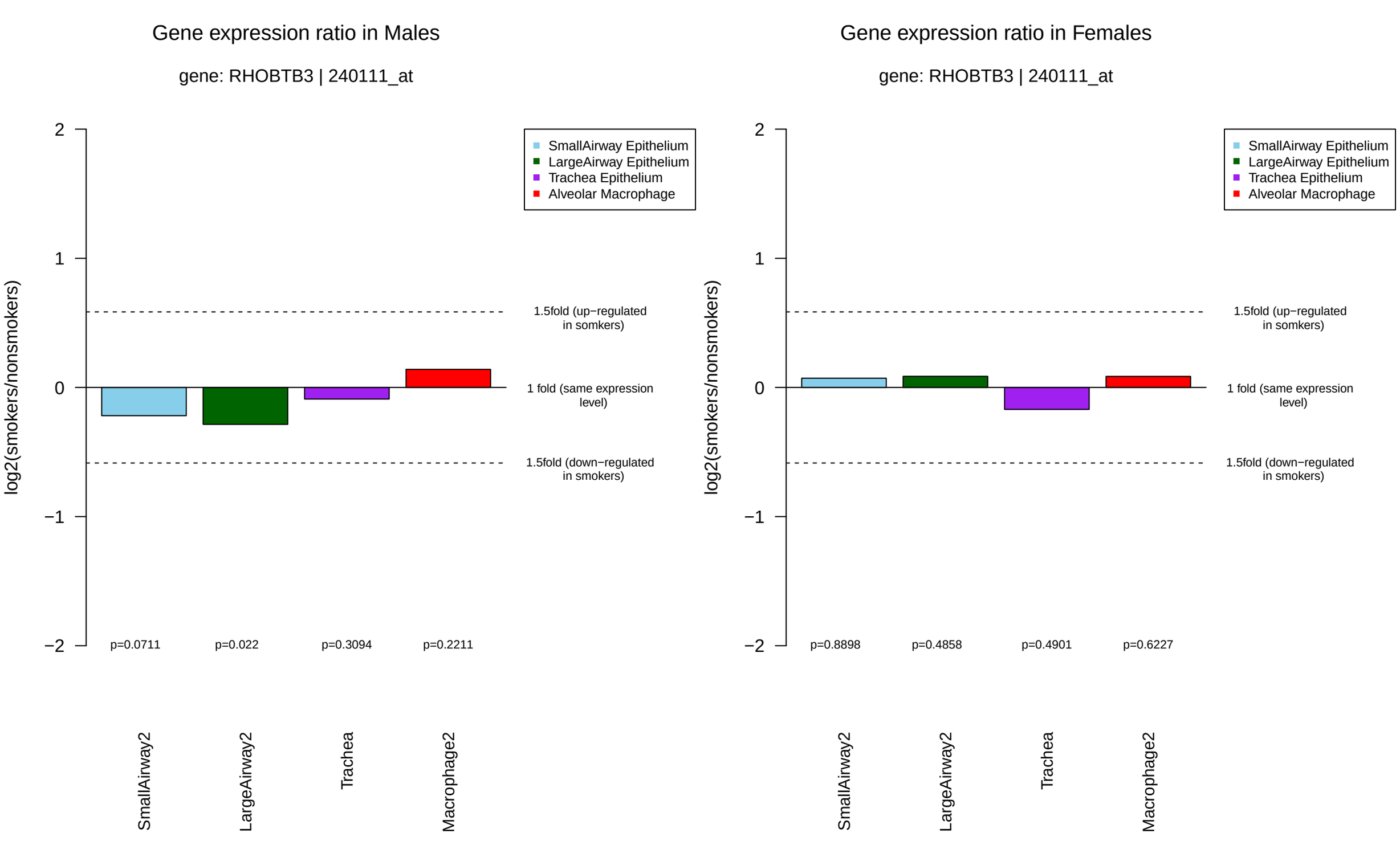
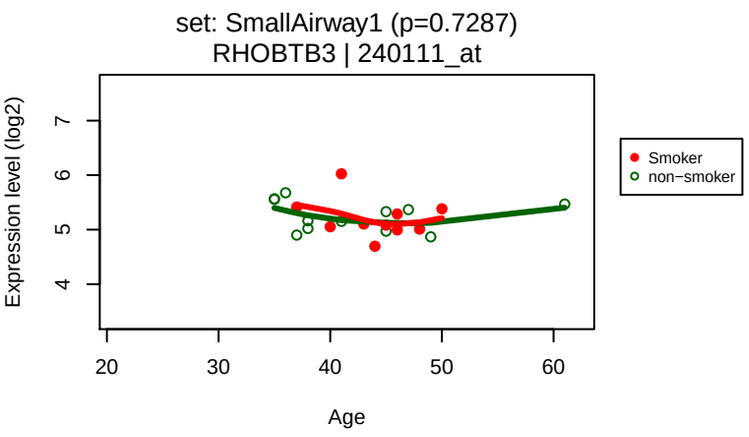
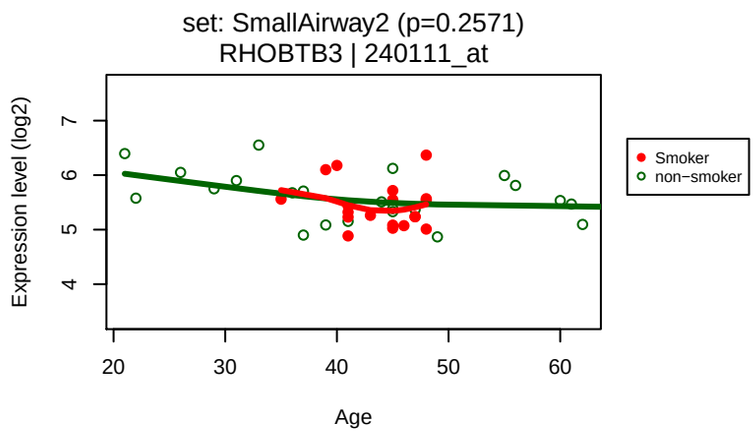
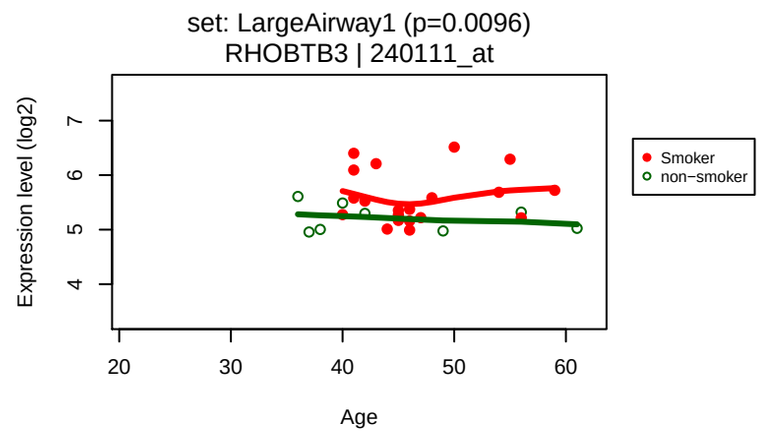

Gene expression of smokers and nonsmokers
gene: RHOBTB3 | 240111_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7287
p=0.2571
p=0.3942
p=0.0096
p=0.0639
p=0.1268
p=0.5074
p=0.182
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

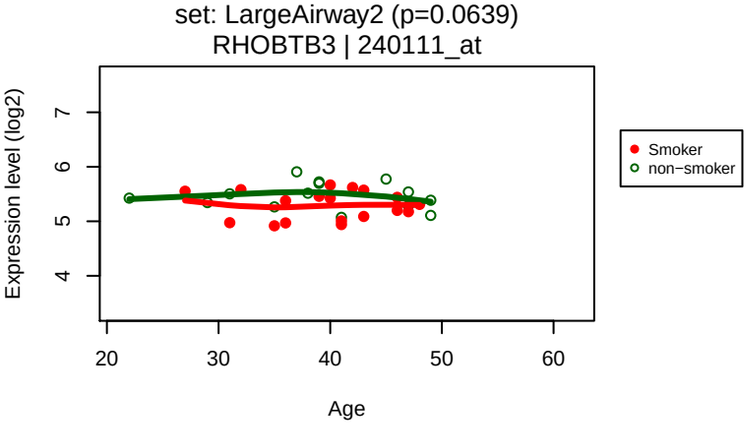

Gene expression ratio (smokers/nonsmokers)
gene: RHOBTB3 | 240111_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7287
p=0.2571
p=0.3942
p=0.0096
p=0.0639
p=0.1268
p=0.5074
p=0.182
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

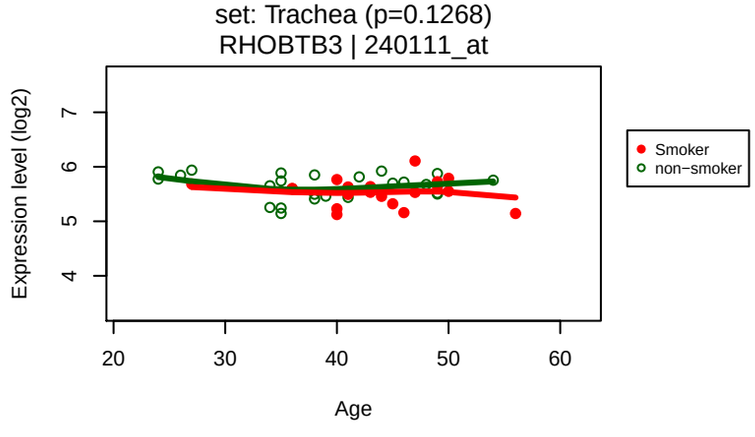

Gene expression ratio in Males
gene: RHOBTB3 | 240111_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0711
p=0.022
p=0.3094
p=0.2211
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RHOBTB3 | 240111_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8898
p=0.4858
p=0.4901
p=0.6227
SmallAirway2
LargeAirway2
Trachea
Macrophage2

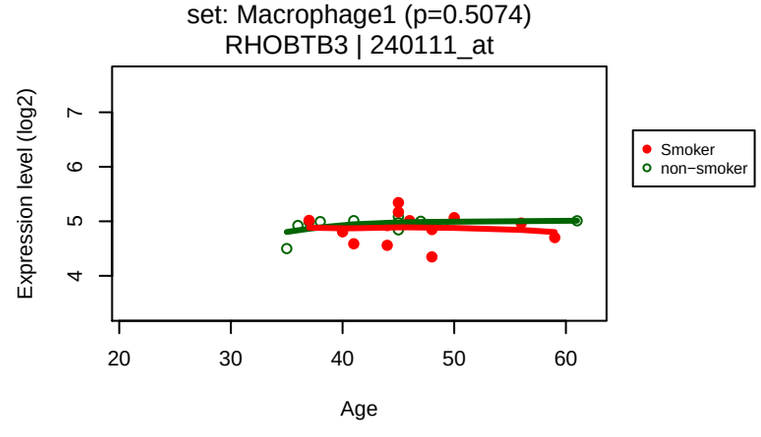

set: SmallAirway1 (p=0.7287)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker

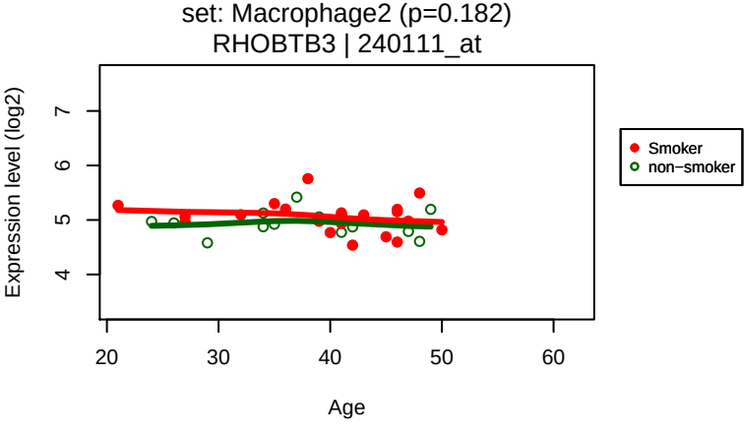

set: SmallAirway2 (p=0.2571)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker

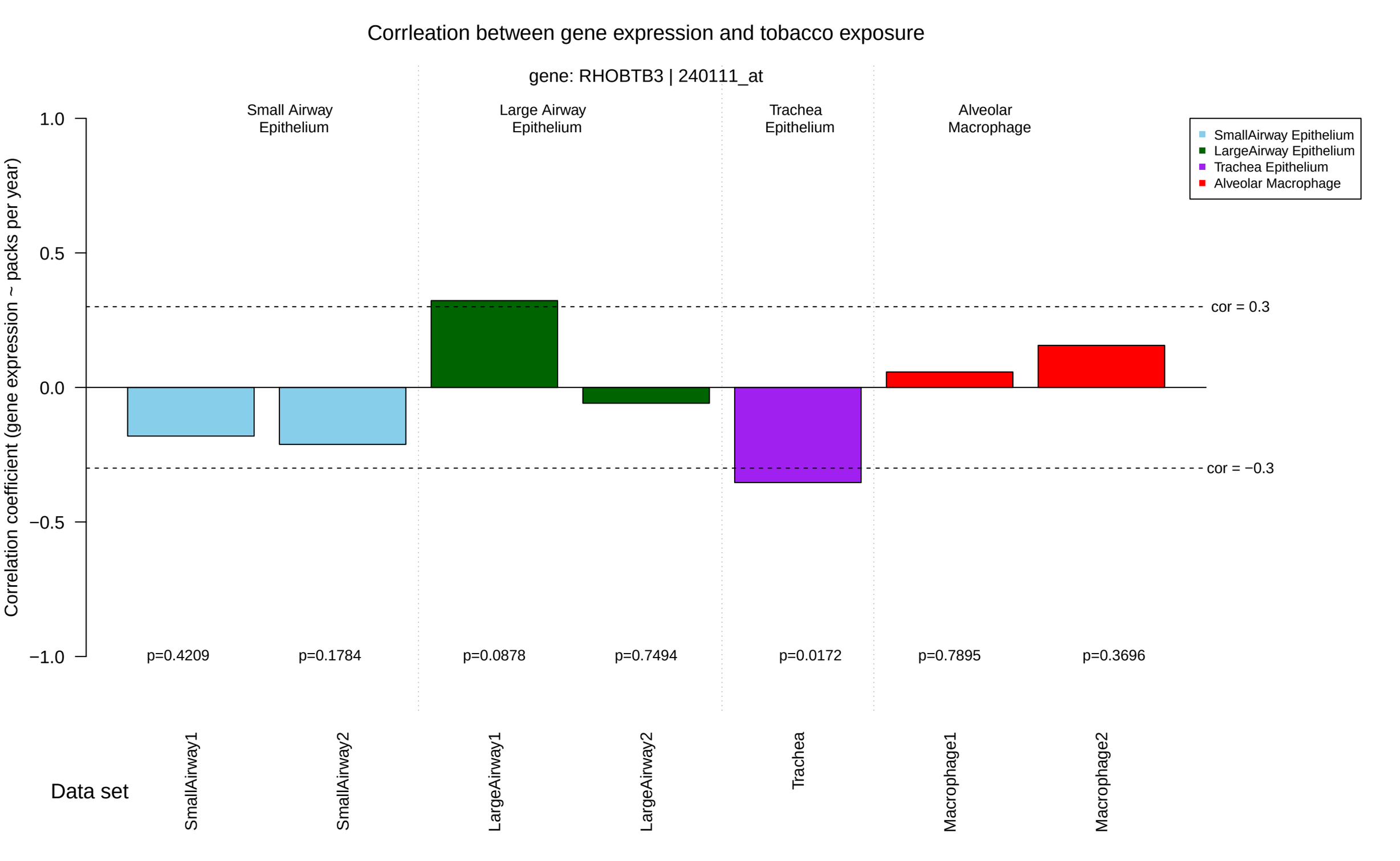

set: LargeAirway1 (p=0.0096)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0639)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1268)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5074)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.182)
RHOBTB3 | 240111_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RHOBTB3 | 240111_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4209
p=0.1784
p=0.0878
p=0.7494
p=0.0172
p=0.7895
p=0.3696
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2